

HOMEWORK 10

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 4.1 - CALCULATION OF RESIDUES

1. Find the residues of the following functions at the indicated points:

(a) $\frac{e^z - 1}{\sin z}, z_0 = 0$

Let $g(z) = e^z - 1$ and $h(z) = \sin z$. Notice that $g(0) = 0$ and $g'(0) = e^0 = 1 \neq 0$. Likewise, $h(0) = 0$ and $h'(0) = \cos 0 = 1$. So, g and h both have $z_0 = 0$ as a zero of order 1 and $\frac{g(z)}{h(z)} = \frac{e^z - 1}{\sin z}$ has $z_0 = 0$ as a removable singularity by Proposition 4.1.1. Hence, $\text{Res}\left(\frac{e^z - 1}{\sin z}; 0\right) = 0$.

(b) $\frac{1}{e^z - 1}, z_0 = 0$

Let $g(z) = 1$ and $h(z) = e^z - 1$. Then, $h(0) = 0$, while $h'(0) = e^0 = 1 \neq 0$. So, g has $z_0 = 0$ as a zero of order 0 and h has $z_0 = 0$ as a zero of order 1. Thus, $\frac{g(z)}{h(z)} = \frac{1}{e^z - 1}$ has $z_0 = 0$ as a pole of order 1 (i.e. a simple pole). By Proposition 4.1.2 $\text{Res}\left(\frac{1}{e^z - 1}; 0\right) = \frac{g(0)}{h'(0)} = 1$.

(c) $\frac{z + 2}{z^2 - 2z}, z_0 = 0$

Let $g(z) = z + 2$ and $h(z) = z^2 - 2z = z(z - 2)$. Then, $g(0) = 2 \neq 0$, while $h(0) = 0$ and $h'(0) = 2(0) - 2 = -2$. So, g has $z_0 = 0$ as a zero of order 0 and h has $z_0 = 0$ as a zero of order 1. Thus, $\frac{g(z)}{h(z)} = \frac{z + 2}{z^2 - 2z}$ has $z_0 = 0$ as a pole of order 1 (i.e. a simple pole). By Proposition 4.1.2 $\text{Res}\left(\frac{z + 2}{z^2 - 2z}; 0\right) = \frac{g(0)}{h'(0)} = -1$.

(d) $\frac{1 + e^z}{z^4}, z_0 = 0$

Let $g(z) = 1 + e^z$ and let $h(z) = z^4$. Then, $g(0) = 2 \neq 0$, while $h(0) = h'(0) = h''(0) = h^{(3)}(0) = 0$ and $h^{(4)}(0) = 24 \neq 0$. So, g has $z_0 = 0$ as a zero of order 0 and h has $z_0 = 0$ as a zero of order 4. Thus, $\frac{g(z)}{h(z)} = \frac{1 + e^z}{z^4}$ has $z_0 = 0$ as a pole of order 4. Since $k = 4$ is the smallest positive integer such that $\lim_{z \rightarrow z_0} (z - z_0)^k \frac{g(z)}{h(z)}$ exists, Proposition 4.1.6 gives that $\text{Res}\left(\frac{1 + e^z}{z^4}; 0\right) = \frac{g^{(4-1)}(0)}{(4-1)!} = \frac{e^0}{3!} = \frac{1}{6}$.

(e) $\frac{e^z}{(z^2 - 1)^2}, z_0 = 1$

Let $g(z) = e^z$ and $h(z) = (z^2 - 1)^2$. Then, $g(1) = e \neq 0$, while $h(1) = h'(1) = h''(1) = h^{(3)}(1) = 0$ and $h^{(4)}(1) = 384$. So, g has $z_0 = 0$ as a zero of order 0 and h has $z_0 = 0$ as a zero of order 4. Thus, $\frac{g(z)}{h(z)} = \frac{e^z}{(z^2 - 1)^2}$ has $z_0 = 0$ as a pole of order 4. Notice that $k = 2$ is the smallest positive integer such that $\lim_{z \rightarrow z_0} (z - z_0)^k \frac{g(z)}{h(z)}$ exists, as evidenced by $\lim_{z \rightarrow z_0} (z - 1)^2 \frac{e^z}{(z^2 - 1)^2} = \lim_{z \rightarrow 1} \frac{e^z}{(z + 1)^2} = \frac{e}{4}$. Therefore, Proposition 4.1.6 gives that $\text{Res}\left(\frac{e^z}{(z^2 - 1)^2}; 1\right) = \frac{\frac{d^{(2-1)}}{dz^{(2-1)}}\left(\frac{e^z}{(z+1)^2}\right)\Big|_{z=1}}{(2-1)!} = \frac{e^z(z-1)}{(z+1)^3}\Big|_{z=1} = 0$.

2. Explain what is wrong with the following reasoning:

Let $f(z) = \frac{1+e^z}{z^2} + \frac{1}{z}$. Since $f(z)$ has a pole at $z = 0$, the residue at that point is the coefficient of $\frac{1}{z}$, namely 1.

Then, compute the residue correctly.

The residue of f at z_0 is defined to be the coefficient b_1 in the Laurent expansion of $f(z)$ around z_0 written as

$$f(z) = \sum_{k=0}^{\infty} a_k(z-z_0)^k + \sum_{k=1}^{\infty} b_k(z-z_0)^{-k}.$$

In this instance, notice that $f(z) = \frac{1+e^z}{z^2} + \frac{1}{z}$ is not written as a Laurent expansion about $z_0 = 0$. So, we cannot be assured that $b_1 = 1$ in the Laurent expansion of f , even though it appears in the numerator of $\frac{1}{z}$ as the function is currently written (which, as noted, is not in the form of a Laurent expansion around $z_0 = 0$). As we will prove in Problem 3, $\text{Res}\left(\frac{e^z-1}{z^2} + \frac{1}{z}; 0\right) = \text{Res}\left(\frac{e^z-1}{z^2}; 0\right) + \text{Res}\left(\frac{1}{z}; 0\right)$, and it may be the case the $\text{Res}\left(\frac{e^z-1}{z^2}; 0\right) \neq 0$. Indeed, notice that around $z = 0$, we have

$$\begin{aligned} e^z &= \sum_{k=0}^{\infty} \frac{z^k}{k!} \\ e^z - 1 &= \sum_{k=1}^{\infty} \frac{z^k}{k!} \\ e^z - 1 &= \sum_{k=0}^{\infty} \frac{z^{k+1}}{(k+1)!} \\ \frac{e^z - 1}{z^2} &= \frac{1}{z^2} \sum_{k=0}^{\infty} \frac{z^{k+1}}{(k+1)!} \\ \frac{e^z - 1}{z^2} &= \sum_{k=0}^{\infty} \frac{z^{k-1}}{(k+1)!} \\ \frac{e^z - 1}{z^2} &= \frac{1}{z} + \sum_{k=1}^{\infty} \frac{z^{k-1}}{(k+1)!} \\ \frac{e^z - 1}{z^2} + \frac{1}{z} &= \frac{2}{z} + \sum_{k=1}^{\infty} \frac{z^{k-1}}{(k+1)!} \end{aligned}$$

So, in this case, $\text{Res}\left(\frac{e^z-1}{z^2} + \frac{1}{z}; 0\right) = 2$.

3. If f_1 and f_2 have residues r_1 and r_2 at z_0 , show that the residue of $f_1 + f_2$ at z_0 is $r_1 + r_2$.

Suppose that f_1 and f_2 have residues r_1 and r_2 at z_0 . By definition, this means that f_1 and f_2 have Laurent expansions around z_0 of the form

$$f_1(z) = \sum_{k=0}^{\infty} a_k(z-z_0)^k + \sum_{k=1}^{\infty} b_k(z-z_0)^{-k}$$
$$f_2(z) = \sum_{k=0}^{\infty} c_k(z-z_0)^k + \sum_{k=1}^{\infty} d_k(z-z_0)^{-k}$$

such that $b_1 = \text{Res}(f_1, z_0)$ and $d_1 = \text{Res}(f_2, z_0)$. Since the Laurent expansions of $f_1 + f_2$ is

$$(f_1 + f_2)(z) = \sum_{k=0}^{\infty} (a_k + c_k)(z-z_0)^k + \sum_{k=1}^{\infty} (b_k + d_k)(z-z_0)^{-k},$$

it follows that $\text{Res}(f_1 + f_2; z_0) = b_1 + d_1 = \text{Res}(f_1; z_0) + \text{Res}(f_2; z_0)$